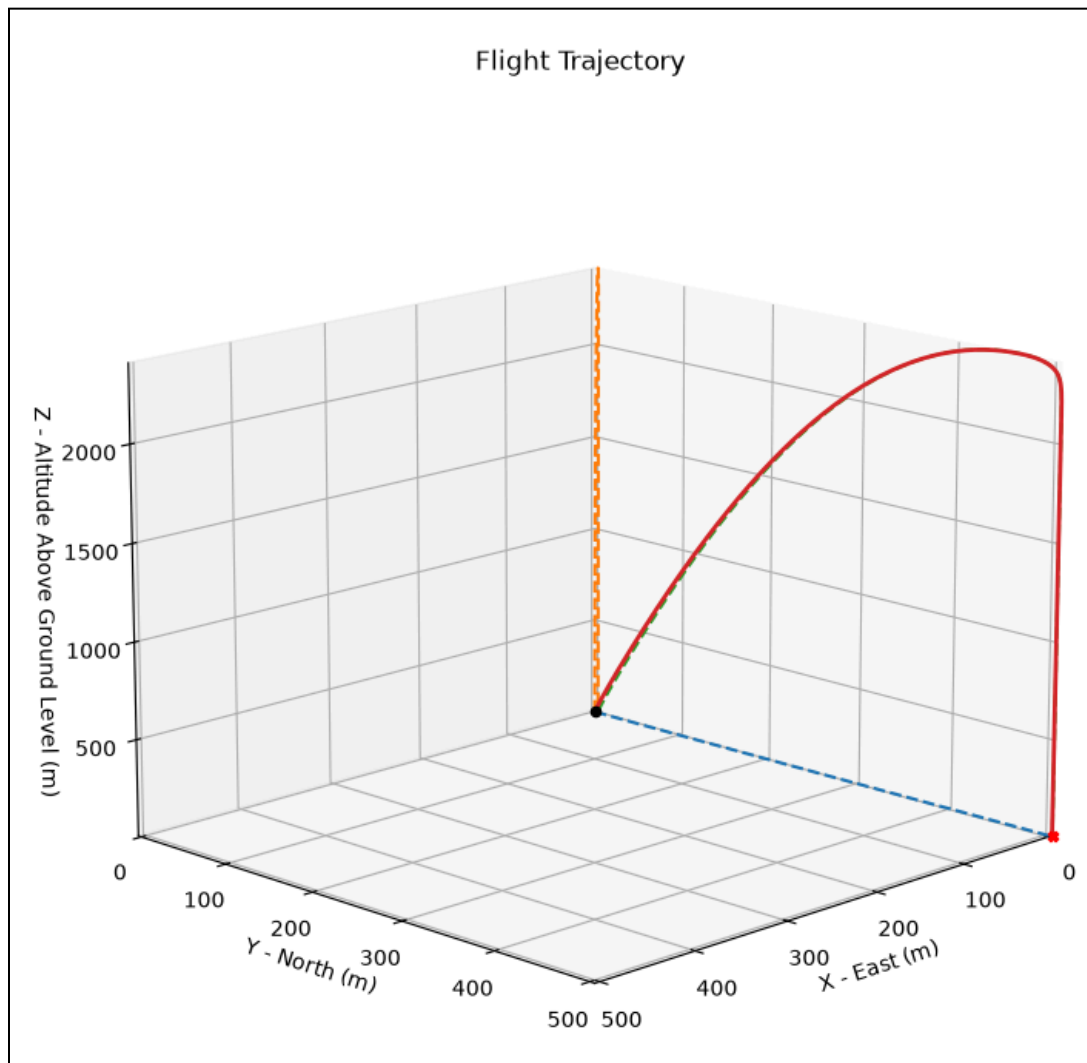
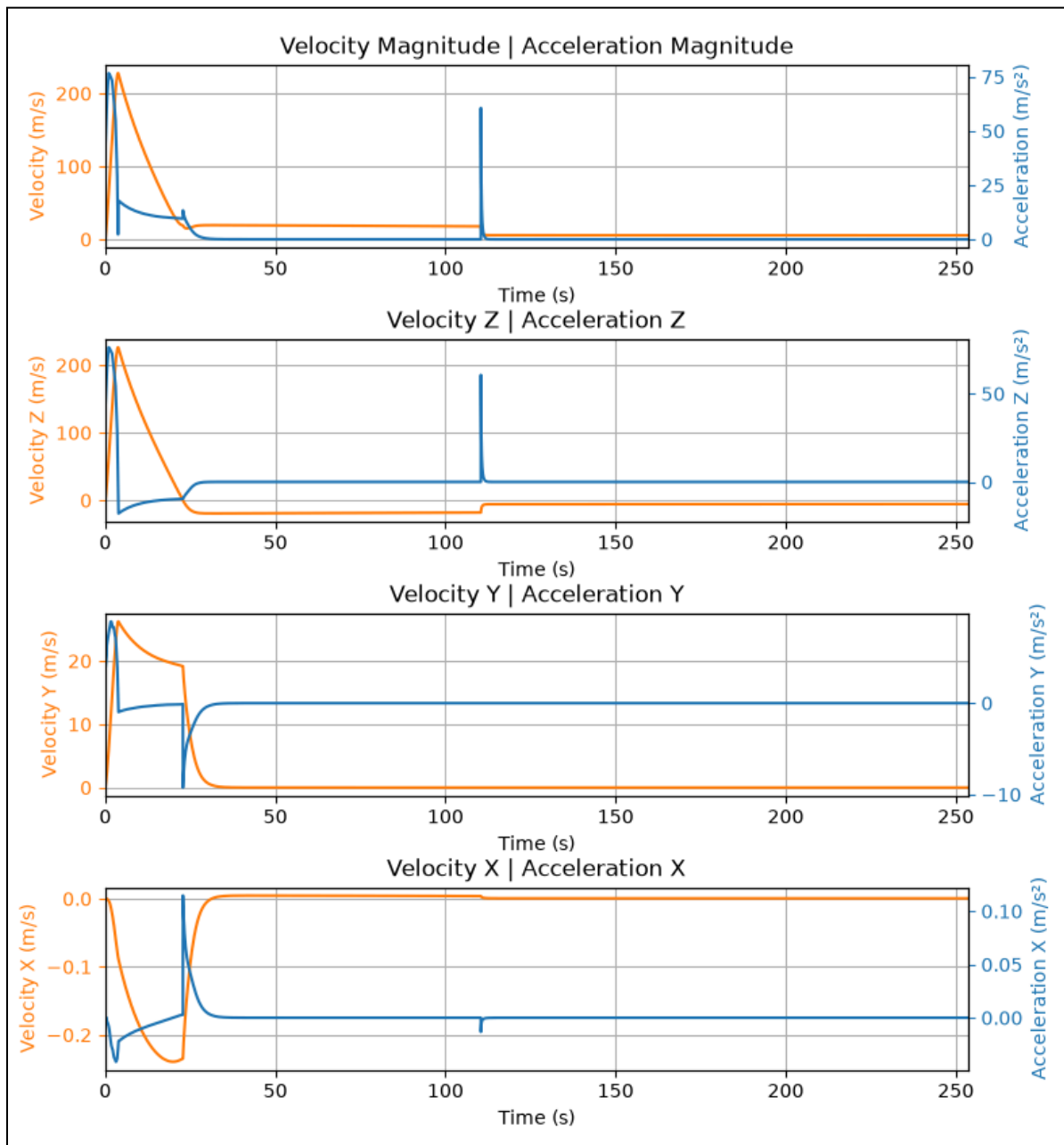


figure 1a



- the curved trajectory (red line) shows the rocket's arched flight path (caused by the 85-degree launch angle), peaking at roughly 3,800 meters
- the descent drift illustrates how far the rocket travels horizontally while being carried by the wind under its parachutes

figure 1b



- the immediate spikes on the left show rapid acceleration from the 3.9-second motor firing
- the sharp blue spikes map the physical "jolt" of the parachutes opening (a small tug at the peak of the flight), and a massive braking force at 800 meters

output

Metric	Value

Max altitude	3808.63 m
Max velocity	227.82 m/s
Burn time	3.90 s
Flight duration	253.52 s

- confirms the exact top speed (227.82 m/s) and highest altitude (3808.63 m)
- shows that a quick 3.9-second engine burn results in a total flight time of just over 4 minutes (253.52 s)